TASK TELANGANA · CLOUD SOFTWARE DEVELOPER TRAINING

Module 03: AWS Core Services & Infrastructure Master Handbook

The 3-Day Technical Deep Dive: Compute, Storage Triad, Databases, VPC Networking, Containers (Docker/ECR/ECS) & IaC CloudFormation

Trainer: Sundeep Hiremath | **Center:** Nalgonda Training Center | **Duration:** 3-Day Course Delivery | **Alignment:** AWS DVA-C02

TRAINER OVERVIEW: This master guide is compiled specifically for the Nalgonda training cohort. It covers the full technical breadth required for enterprise cloud engineering and AWS Certified Developer (DVA-C02) standards. Concepts are grounded in **The Grand Vehicle Metaphor** so students can quickly bridge real-world hardware intuition to enterprise AWS services.

1. The Grand Vehicle Metaphor: The Cloud as an Automobile Fleet

To eliminate confusion among dozens of cloud services, think of AWS as an on-demand vehicle rental and logistics agency:

AWS Service	Vehicle Metaphor	Real-World Function
Amazon EC2	**The Car Engine & Chassis**	Rent an engine on-demand. Choose horsepower (Instance Type: *t3.micro*), model (AMI: *Amazon Linux*), and ignition key (Key Pair *.pem*).
Amazon EBS	**The Car's Personal Trunk**	Local hard drive bolted to 1 car chassis in a parking garage (AZ). Data stays safe when engine turns off.
Amazon EFS	**The Shared Fleet Roof Cargo**	Shared roof rack accessed simultaneously by 100+ cars across multiple garages via NFS protocol.
Amazon S3	**The Global Container Depot**	Infinite worldwide warehouse outside the vehicle. Store objects in buckets; reachable via web URLs (HTTPS). 11 9's durability.
Amazon VPC	**The Gated Industrial Campus**	Private gated corporate park with secure perimeter walls. Only authorized vehicles enter.
Internet Gateway	**Main Highway Gate**	Two-way highway opening connecting public showroom lanes to the internet.
NAT Gateway	**One-Way Security Escort**	Allows private vault cars to fetch fuel/software without letting outside internet cars in.
Security Group	**Car's Personal Bodyguard**	**Stateful:** If a passenger is approved to enter, they can step back out automatically.
Network ACL	**Highway Border Toll Booth**	**Stateless:** Checks license plates on entry AND explicitly checks license plates on exit.

2. Global Infrastructure: Regions, Availability Zones & Multi-AZ

- **AWS Region:** A distinct geographic location (e.g. ap-south-1 Mumbai, us-east-1 N. Virginia). Criteria to choose: 1) Latency to end-users, 2) Data sovereignty/compliance laws, 3) Service feature availability, 4) Regional cost variations.

- **Availability Zone (AZ):** One or more isolated data centers with independent power and networking within a Region (e.g. ap-south-1a, 1b, 1c). Connected by private sub-millisecond optical fiber.
- **Multi-AZ Architecture:** Deploying redundant resources across 2 or more AZs in the same region. Features **synchronous replication**: if one data center floods, automatic failover shifts traffic to the standby AZ in seconds with zero data loss (99.99% availability).
- **Multi-Region Architecture:** Duplicating resources across continents (e.g. Mumbai + Frankfurt) with asynchronous replication for ultimate disaster recovery.

3. Amazon EC2 & The Storage Triad (EBS, EFS, S3)

- **EC2 Lifecycle:** *Pending* ➔ *Running* ➔ *Stopping* ➔ *Stopped* (Compute billing drops to $0.00/hr, storage remains billed, public IP is released) ➔ *Terminated* (Permanently destroyed).
- **Storage Shootout: EBS** is local block storage bound to 1 AZ for OS and databases. **EFS** is a multi-AZ network file system mounted by 100+ servers concurrently via NFS. **S3** is an infinite global object store accessed via REST API with 11 9's durability.
- **S3 Security:** Always keep **Block Public Access (BPA)** ON. Attach an HTTPS-enforced bucket policy to prevent unencrypted transmissions.

4. Databases: RDS vs Aurora vs DynamoDB vs ElastiCache (Redis)

AWS provides specialized database engines tailored for specific data access patterns:

Database	Model	Key Architectural Feature	Ideal Use Case
Amazon RDS	Relational SQL	Managed Postgres/MySQL with Multi-AZ automated synchronous standby failover.	Transactional enterprise ERP, financial ledgers, legacy SQL schemas.
Amazon Aurora	Relational SQL	6-way distributed replication across 3 AZs. 5x faster than standard MySQL. Serverless v2.	High-throughput enterprise web applications and auto-scaling APIs.
Amazon DynamoDB	NoSQL Key-Val	Serverless, single-digit millisecond latency at any scale. Partition Key + Sort Key.	E-commerce carts, user profiles, gaming leaderboards, IoT sensor telemetry.
Amazon ElastiCache	In-Memory (Redis)	Sub-millisecond read/write by caching frequently requested data in RAM.	Web session caches, real-time analytics counters, database read-offloading.

5. VPC Networking, Edge Delivery & Perimeter Security

- **VPC & Subnets:** Standard VPC CIDR 10.0.0.0/16 (65,536 private IPs). **A subnet is Public ONLY if its route table points 0.0.0.0/0 to an Internet Gateway (IGW).** Private subnets route outbound traffic via a **NAT Gateway** residing in a public subnet.
- **Security Groups vs NACLs:** Security Groups are **stateful** and operate at the instance level (allow rules only). Network ACLs are **stateless** and operate at the subnet boundary (allow and deny rules, evaluated in numerical order).
- **ALB vs NLB:** Application Load Balancer operates at Layer 7 (HTTP/HTTPS URL path routing). Network Load Balancer operates at Layer 4 (ultra-low latency TCP/UDP millions of requests/sec).
- **CloudFront & AWS WAF:** CloudFront caches static media at 600+ edge locations. AWS WAF inspects Layer 7 traffic to block SQL Injection, XSS, and rate-limits brute force attacks.

6. Containers: Docker, Amazon ECR & Amazon ECS vs AWS Lambda

A Docker container packages code and dependencies into a lightweight unit that shares the host OS kernel, starting in seconds compared to heavy virtual machines.

6.1 Pushing Docker Images to Amazon ECR via AWS CLI

```
# 1. Authenticate Docker CLI to your private Amazon ECR registry
aws ecr get-login-password --region ap-south-1 | docker login --username AWS --password-stdin
123456789012.dkr.ecr.ap-south-1.amazonaws.com
# 2. Build your local Docker image
docker build -t task-cloud-app:latest .
# 3. Tag image with ECR repository URI
docker tag task-cloud-app:latest 123456789012.dkr.ecr.ap-south-1.amazonaws.com/task-cloud-app:latest
# 4. Push to Amazon ECR
docker push 123456789012.dkr.ecr.ap-south-1.amazonaws.com/task-cloud-app:latest
```

6.2 Architecture Showdown: Amazon ECS vs AWS Lambda Container Images

Architectural Metric	Amazon ECS (Fargate / EC2)	AWS Lambda Container Images
Execution Timeout	**Unlimited / Indefinite** (Runs 24/7/365)	**Strict 15-Minute Maximum**
Compute Model	Persistent web services, background daemons	Event-driven, serverless execution (scale-to-zero)
Max Image Size	Dozens of Gigabytes supported	Maximum 10 GB container image
Billing	Per-second of allocated vCPU & RAM	Per-millisecond of actual execution time ($0 if idle)
Best Use Case	Continuous REST APIs, web apps, message queues	S3 image processing, webhooks, scheduled cron ETL

7. Infrastructure as Code (IaC) & CloudFormation Drift Detection

Infrastructure as Code (IaC) replaces error-prone manual console clicking with version-controlled YAML/JSON templates. CloudFormation manages the full lifecycle of your stacks, providing automatic rollback on failure and preview Change Sets.

7.1 Understanding & Catching Infrastructure Drift

WHAT IS DRIFT? Drift occurs when an engineer makes out-of-band manual changes in the AWS Console (e.g. changing an EC2 instance type from t3.micro to m5.large or opening Port 22 in a Security Group) without modifying the CloudFormation template.

HOW TO DETECT & REMEDIATE DRIFT:
1. Run **Detect stack drift** in the CloudFormation Console (or CLI: aws cloudformation detect-stack-drift --stack-name <name>).
2. Review the drift report: AWS flags drifted properties in red side-by-side with template expected values.
3. **Remediation:** Either revert the manual console change back to the template baseline, OR update the CloudFormation YAML template to adopt the new configuration and deploy via a Change Set.

8. Student Zero-Dollar Sunset Checklist

1. Stop or Terminate all practice EC2 instances.
2. Delete orphan EBS volumes in 'available' status.
3. Delete custom CloudFormation stacks (automatically cleans up all child resources!).
4. Release any unallocated Elastic IPs.
5. Check AWS Billing Dashboard to verify $0.00 balance.